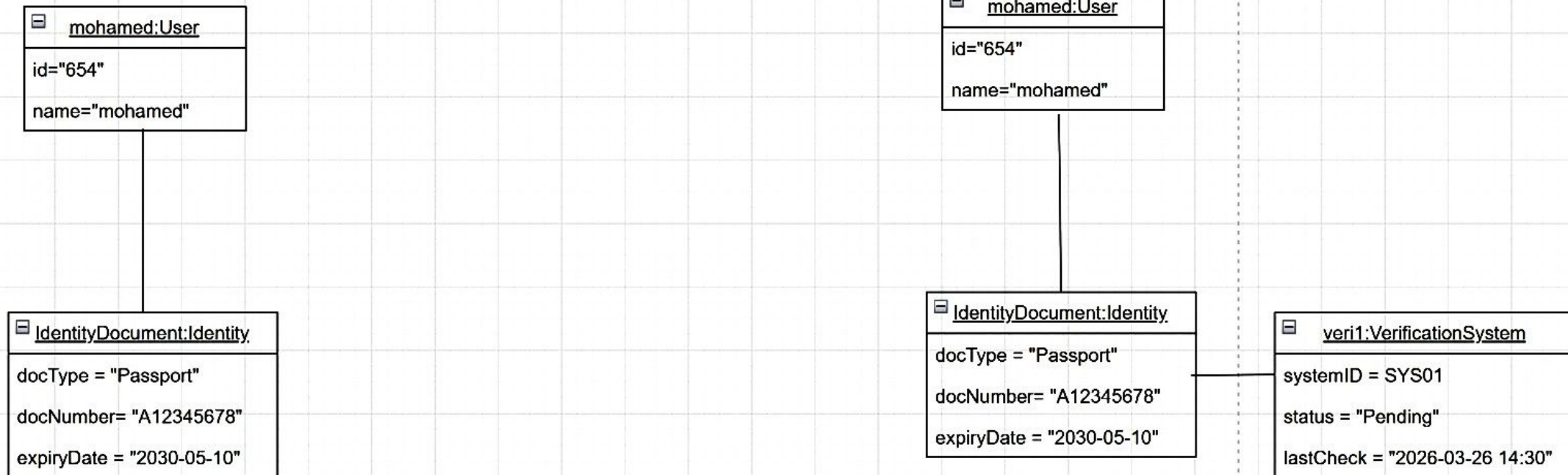


object diagram for User Identity Verification (KYC)



pre-condition:

User must have a valid account in the system.

post-condition:

User's identity is verified successfully, or flagged for manual review.

Verification status is stored in the system.

Role-Based access control (RBAC)

 <u>user1:User</u>
id="453"
name="mohamed"
type="sitter"

 <u>mohamed:ServiceProvider</u>
type="sitter"

pre-condition:

the system knows the user's role

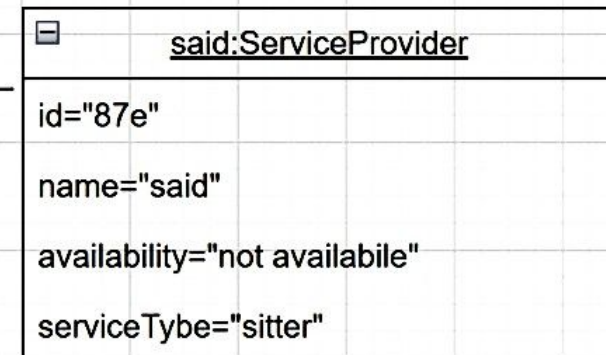
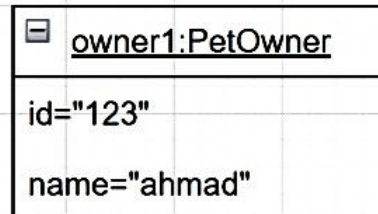
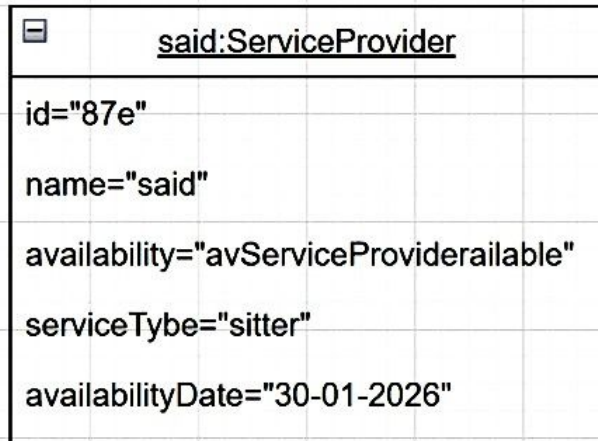
 <u>user1:User</u>
id="453"
name="mohamed"
type="sitter"
permission="view schedule"

 <u>mohamed:ServiceProvider</u>
type="sitter"
availabilityDate="02_9_2026"

post-condition:

access is allowed if permitted; otherwise, it is denied

Sitter Availability Conflict Resolver

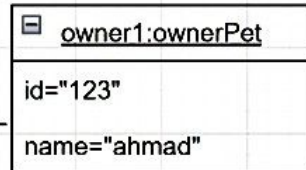
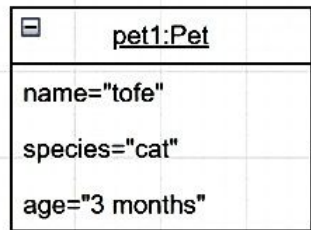


pre-condition:

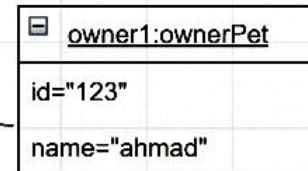
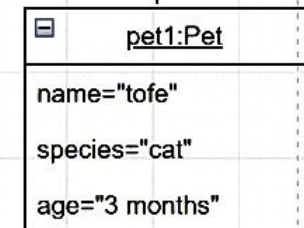
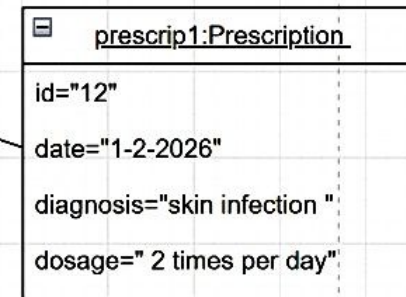
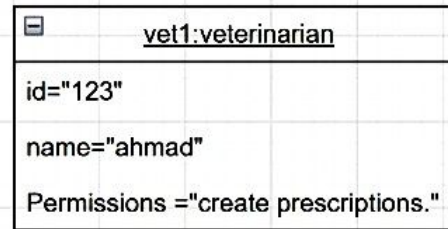
the sitter must have an availability schedule and the new booking has a date and time.

post-condition:

if there's a conflict, the system alerts the sitter and owner and suggests rescheduling; if no conflict, the booking is added successfully.



Digital prescription engine



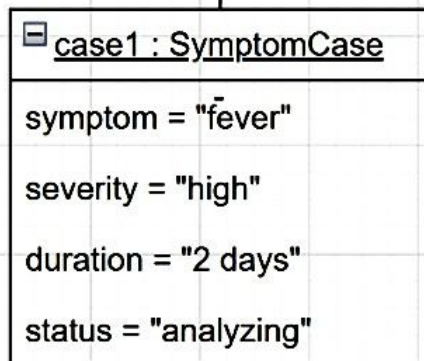
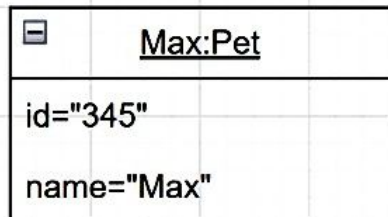
Pre-condition:

the veterinarian must be logged into the system, the pet and owner information must already exist, and the veterinarian must have permission to create prescriptions..

Post-condition:

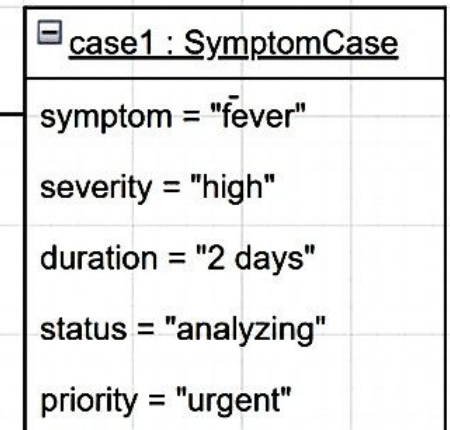
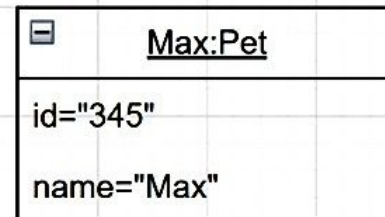
new digital prescription is created and saved in the system, it is linked to the correct pet, includes medication and dosage details, and can be viewed by the pet owner.

Symptom-to-Specialist Triage



pre-condition:

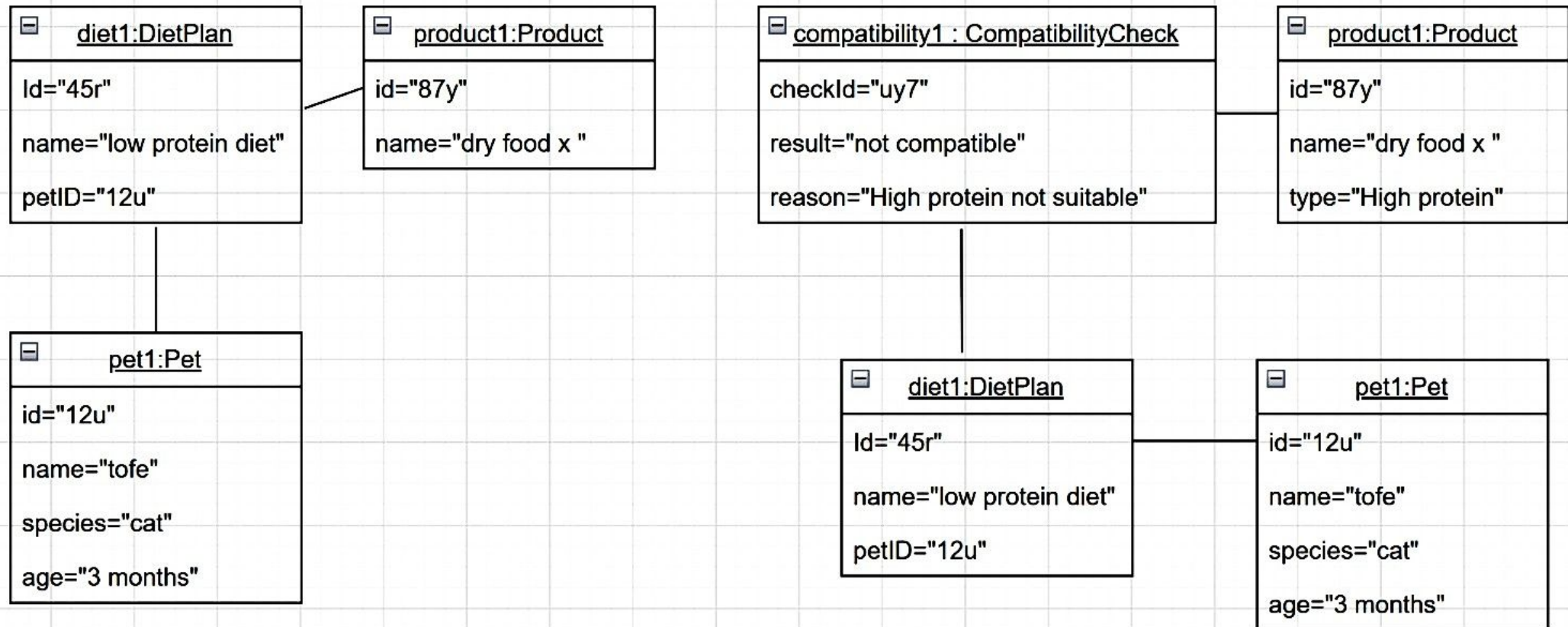
the system receives the pet's symptoms, severity, and duration.



post-condition:

the system selects the most suitable available specialist and sets the case priority.

Medical-Diet Compatibility



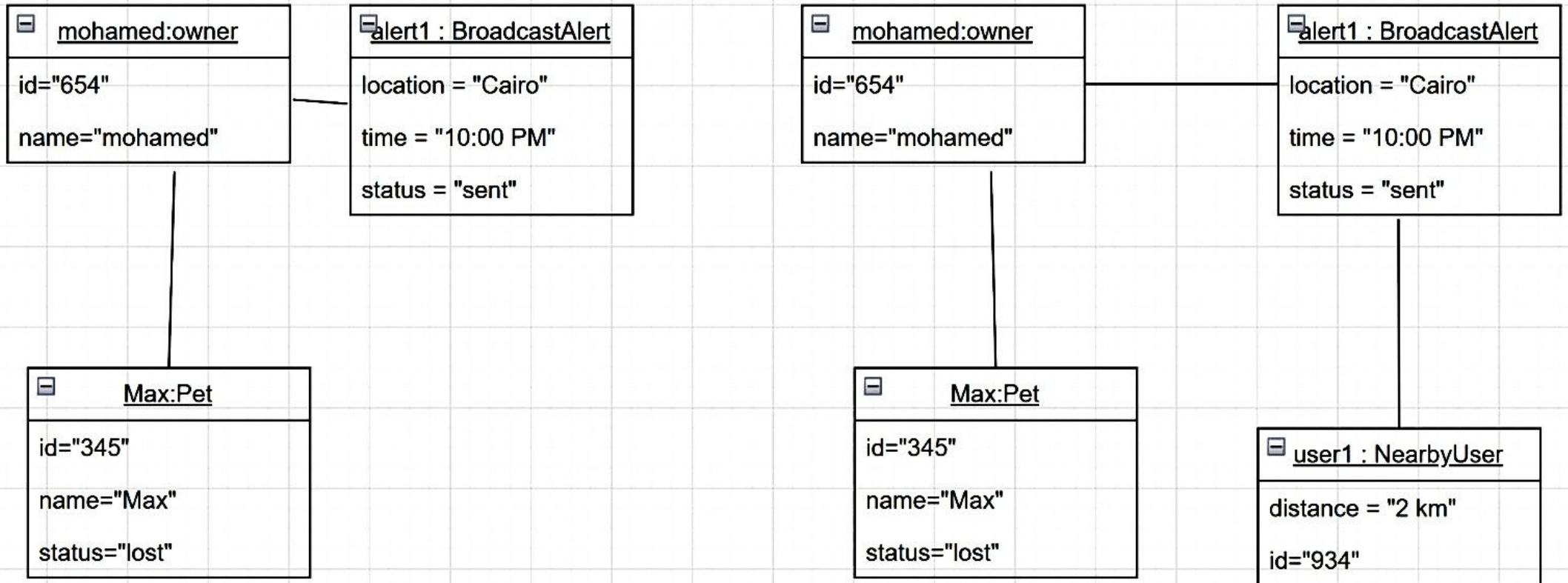
pre-condition:

the pet and its medical condition must be recorded in the system, and the diet plan and product information must already exist

post-condition:

the system checks the compatibility between the pet's condition and the selected product, determines if it is suitable or not, and stores the result with a clear reason.

Emergency Lost Pet Broadcast



pre-condition:

the pet is marked as lost with location details

post-condition:

the system sends alerts to nearby users to help find the pet

Automated Vaccination Scheduler

